

while flexibly utilising additional computing power required for chip design and verification.

Cool, right? Yes, but in case you are wondering, HPCaaS has extended its reach beyond such exotic applications to the relatively mundane. A leading private sector bank in India has opted for HPCaaS for algorithms such as quick loan approvals and fraud detection in loans to industry. CARE Ratings uses HPCaaS for data analytics projects to develop sentiment analysis and predictive models for its ratings business. “As neural network models recommend GPU-based processing, we use HPCaaS in the development of neural network-based deep learning models,” says Tarun Banger, Head-Data Science & Analytics, CARE Ratings. “The high-performance cloud-based environment has helped us develop a scaleable framework for building AI/ML models, ultimately helping us in distributing workloads seamlessly. Cloud computing has enabled us to automate complex, repetitive, high-processing tasks without any human intervention.”

Som Satsangi, SVP and Managing Director of HPCaaS vendor Hewlett Packard Enterprise, India, says: “In the recent past, there has been a steady shift of HPC to the cloud, which has been the fastest-growing part of the HPC market. It becomes difficult for enterprises to manage infrastructure because of the rate and pace of change in technology and the skills needed to operate on-premises HPC. So, instead of buying technology, they are looking to use it as a cloud service.”

Technology consulting companies, too, are leveraging HPCaaS. Tata Elxsi, which provides technology consulting, product design and development services to organisations globally, is leveraging a cloud-based HPC solution as it found it far more feasible in the context of service to clients, sales closures and timely deliveries, apart from other usual benefits of a cloud model. “The system goes through many upgrades over the years as technology evolves, resulting in additional costs incurred every few years to maintain an up-to-date physical infrastructure. The on-cloud HPC on (Microsoft’s) Azure was a much better option,” says Sagar MSV, the company’s Manager-Transportation Design & Engineering.

Microsoft is making its HPC services available through the Azure portal and marketplace—

Azure Cycle Cloud and Azure Batch. “These services are free to use, and customers pay for the underlying compute to use with these services. The overall cost depends on the customer use case, workload, and the time duration of running those workloads on Azure,” explains Shivir Chordia, Azure Business Group Lead, Microsoft India. In India, Microsoft says it has improved time to market with faster solve times for a large engineering services organisation, enabled a large engineering university to run molecular simulation workload, and enabled researchers in a large organisation to focus on innovation and research, taking away the overheads of running HPC clusters on-prem.

HPCaaS is proving to be a boon for enterprises and start-ups relying heavily on AI and ML. Orbo.ai, an AI and computer vision-based platform focussed on image enhancement and automation, has to perform extensive AI training over huge image and video datasets. This training requires compute-efficient systems that can run for days without any interruption. Currently, the company is training AI for hair segmentation, face landmarks, signature matching and foetus scan enhancement on Yotta’s HPC virtual machines. “We have been using HPCaaS since the past seven months, and we are deriving two direct benefits from it—uninterrupted system availability for our AI research team, and cost savings,” says Md Danish Jamil, Co-Founder and Technical Lead, Orbo.ai.

As HPCaaS is still nascent in India, players in this space are experimenting with new ways to bring awareness and increase adoption. Bengaluru-based NxtGen Datacen-

Industry Possibilities

HPCaaS can be used across multiple industries. Here’s a sampling

- ➔ **BFSl:** Load approval, detecting abnormalities and intrusion attempts in real time
- ➔ **Healthcare:** Computational power can help process acceleration and reduce discovery time in research and drug discovery
- ➔ **Smart Energy Grids:** Can provide enormous computing power to optimise and maintain energy efficiency in real time
- ➔ **Manufacturing:** Can help in quickly analysing data and provide autonomous capability
- ➔ **Education:** Help in research and innovation